

The fair reduced signature has exponential Hankel rank

Algebraic diagnostic; two independent mathematical reviews completed

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Fix positive counts $c_1, \dots, c_n$. Regard the reduced fair signature as an ordinary noncommutative polynomial by setting

$$F(w) = \begin{cases} \frac{\prod_{i \in S(w)} c_i}{|w|!}, & \text{if every letter in } w \text{ is distinct,} \\ 0, & \text{otherwise.} \end{cases}$$

The empty word has coefficient one. Its Hankel matrix is the matrix $\mathcal{H} = (F(uv))_{u,v}$ indexed by all finite words.

Proposition 1. *Over $\mathbb{R}$, $\text{rank } \mathcal{H} = 2^n$.*

Proof. If u repeats a letter, its whole row is zero. Otherwise the row depends only on its support $S(u)$: the entry is zero if the column repeats a letter or its support meets $S(u)$, and otherwise depends only on the union of their supports. There are 2^n possible supports, proving the upper bound.

For each subset S choose one word listing its elements. Restrict rows to these words and columns to words listing T^c , for all subsets T . The entry is nonzero exactly when $S \subseteq T$. Order both index sets by increasing cardinality, with identical orders within each cardinality. This matrix is upper triangular, and its diagonal entries are all

$$\frac{\prod_{i=1}^n c_i}{n!} > 0.$$

Thus its rank is 2^n . □

There is also a direct positive weighted automaton with 2^n states. A state records the set S of letters already read. Reading $i \notin S$ moves to $S \cup \{i\}$ with weight $c_i/(|S| + 1)$; repeated letters have no transition. Start at the empty set and give every state terminal weight one. The product of transition weights is exactly $F(w)$.

Proposition 2 (A still smaller ordinary completion). *Let $H = \sum_i c_i X_i$ and define the ordinary noncommutative polynomial $E = \sum_{j=0}^n H^j/j!$, without setting repeated-letter coefficients to zero. It agrees with all prescribed squarefree fair coefficients, has nonnegative coefficients, and has Hankel rank $n + 1$.*

Proof. For a word u of length at most n , divide its Hankel row by the positive product of its letter weights, with repetitions counted. The resulting row depends only on $|u|$. Longer rows are zero, so rank is at most $n + 1$. Choose rows X_1^a and columns X_1^b for $0 \leq a, b \leq n$. After positive row and column rescaling, the entries are $1/(a + b)!$ if $a + b \leq n$, and zero otherwise. Reverse the order of the columns. The resulting triangular matrix has diagonal entries $1/n!$, proving the lower bound. □

Scope. These are representations of coefficient data, not constructions of fair dice. A positive axis word has uncontrolled repeated-letter coefficients before passage to the reduced algebra; setting these to zero changes its ordinary Hankel matrix. Consequently no face-count upper or lower bound follows merely by comparing this rank with the length of a word. Nor does the result bound the rank of the endpoint *differential*, whose reduced Lie target has factorial dimension. The reduced target has an exponential-size prefix–suffix linear representation, whereas its squarefree constraints admit an ordinary nonnegative completion of rank only $n + 1$. The latter is the truncated signature of a simultaneous straight path, not of a finite positive axis word. Thus positivity and the unspecified repeated-letter data cannot be discarded when seeking a factorial obstruction by prefix–suffix moment rank.